



Graduation Project Proposal

Project Information		
Project Title	Department/Faculty/University	Project Field/Discipline
Qurany	computer science/ software engineering/ Cairo university	Software Engineering and Artificial Intelligence
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If the project is sponsored by or initiated by an ICT company, please state its name:



Motivation

Please write why you chose this project idea, explaining clearly the problem that the project is addressing

The motivation behind developing Qurani stems from a real personal and practical need experienced during the process of memorizing and reviewing the Holy Qur'an. As a learner, I aimed to regularly recite large portions of the Qur'an to qualified teachers (Sheikhs) in order to receive accurate feedback and correction. However, due to their limited availability and busy schedules, it became difficult to maintain consistent listening and evaluation sessions.

In an attempt to find alternatives, I searched for knowledgeable friends and qualified reciters who could listen and provide feedback, but this solution was not always reliable or accessible. This challenge led me to explore digital solutions, including the Tarteel AI application, which offers AI-based Qur'an recitation analysis. While the application provided a relatively good experience, it lacked many important features, had limited flexibility, and restricted most of its advanced capabilities behind a paid subscription.

Although the free trial experience was unique and impressive, it raised an important question: *Why is such a valuable service not freely available with broader features, multiple recitation styles (Qirā'āt), and a more comprehensive Qur'anic learning experience?*

From this question, the idea of Qurani was born—to develop a free, inclusive, and more feature-rich Qur'an application that supports learners, memorizers, and reciters by combining advanced technology with accessibility, aiming to serve the Qur'an and its learners in a more comprehensive and inclusive way.

Project Description

Provide a brief description of the project in no more than 200 words.

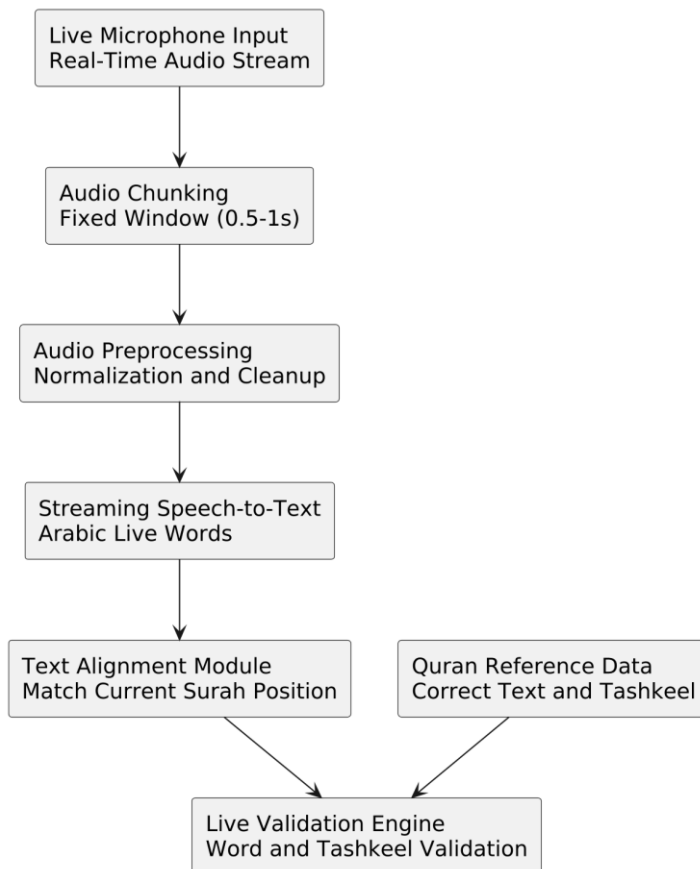
This project aims to develop an Islamic application that helps users improve their Quran recitation. The main focus of the app is that the user recites verses aloud, and the system listens to the recitation, recognizes the verse, and displays the correct text. The app then compares the user's recitation with the original verse and highlights mistakes or differences. This allows users to practice memorization, check their accuracy, and receive guidance on improving their recitation.

In addition to recitation correction, the app will also offer several helpful Islamic features. These may include reading the full Quran, searching for Surahs and Ayahs, saving bookmarks, showing prayer times, Azan notifications, Qibla direction, and daily Islamic reminders such as Duas and Hadiths. The final platform of the application, whether web-based or mobile-based, will be decided later depending on project requirements and available tools.

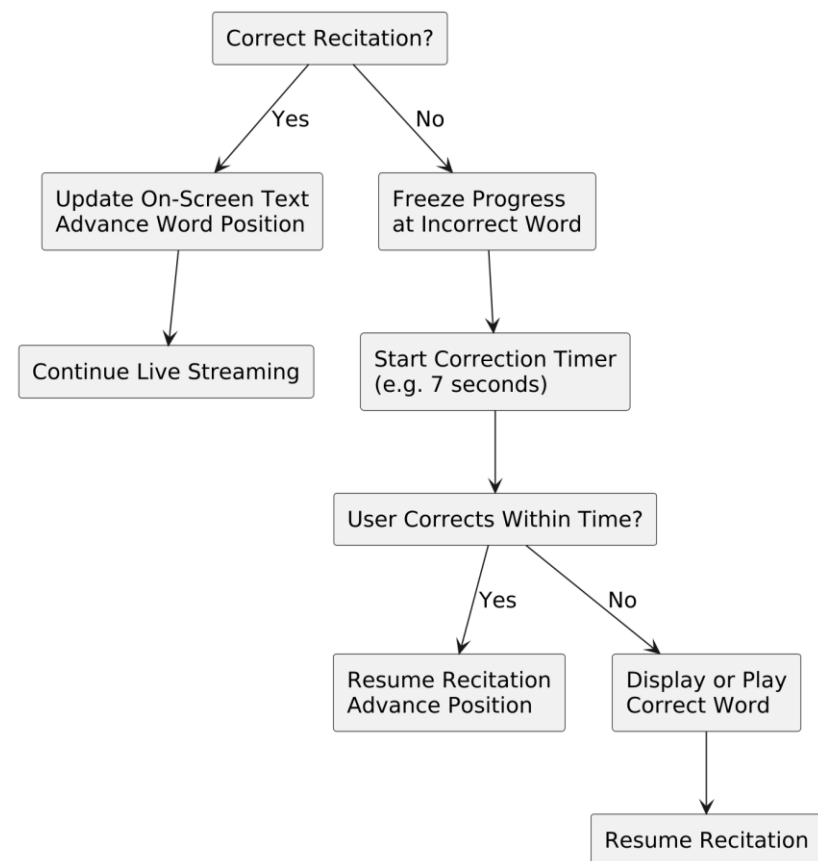
Block Diagram

Please insert the expected or envisioned project detailed block diagram below. This diagram should reflect the main workflow of the system, including its inputs, how they progress through the system, and the expected outputs.

Qurani System - Live Recitation Streaming Flow



Qurani System - Error Handling and Correction Flow





Project Plan

Please define the approach and phases to deliver the intended project outcome.

The project started with an initial research phase focusing on Quran learning applications, particularly recitation recognition and correction systems. Existing solutions and speech-to-text technologies were studied to understand system requirements and feasibility.

After completing the research, detailed documentation was prepared to define the project idea, objectives, key features, and system architecture. This step ensured a clear understanding of the application scope before implementation.

An experimental prototype was then developed using Python to test Quran recitation transcription from audio files, validating the feasibility of speech recognition for this use case.

Based on the prototype results, the next phase involves developing the full live recitation module with real-time verse recognition and mistake highlighting. Additional features such as a Quran reader, bookmarks, prayer times, Azan notifications, Qibla direction, and daily Islamic content will be integrated. The project will conclude with testing, refinement, and final deployment as a graduation project.

Prototype Prospects

List the Egyptian ICT companies or beneficiaries that may be interested in the developed prototype and the end-users/customers (name the specific class of individuals, governmental agencies, ministries ... etc. that will benefit from the prototype)

List of ICT Companies and/or Beneficiaries:

- Ministry of Communications and Information Technology (MCIT) – government body promoting digital Egypt and ICT innovation; could support or adopt the app for national educational projects.
- Ministry of Awqaf & Al-Azhar Al-Sharif – key religious and educational authorities focused on Quran teaching and recitation, likely beneficiaries of a Quran recitation and correction application. (context from religious education focus)
- Islamic education institutions and Quran memorization centers – schools and learning centers that teach Tajweed and memorization techniques would benefit directly from technology that supports recitation correction.

Potential End-Users/ Consumers:

- Students and learners of Quran / general Muslim population – individuals who want to improve recitation accuracy and memorization with guided feedback. But must be already a little familiar with the recitation and tajweed
- Parents and tutors – those supporting children or learners in Quran study who would use the app regularly for practice and tracking progress.